Timing a possible encounter

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Separate orbital period from an encounter time.

SUCCESS CRITERIA - I can...

1. Use a clearly stated period and time reference.
2. Check positions of both bodies at the same time.
3. Explain why geometry and uncertainty matter.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “encounter” and “period” to explain your model or result. Sentence starters are optional.

encounter: objects reaching nearby positions at the same time

period: time for one complete cycle or orbit

prediction: what a model says may happen next

01 Bring back an earlier idea

A car follows a circle at steady 5 m/s.
Sketch velocity at two points. Could net force be zero while it follows this path?

Word help: net force = the vector sum of all forces on one object

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $t_{arrival} = t_{start}$ + travel time.
An encounter needs matching position and time. A period T is one complete orbit. Use the supplied fictional timing model.

Worked reasoning: Two objects pass the same crossing point: A at months 2, 8, 14 and B at months 5, 14, 23. Only month 14 is a shared time in these lists.

Your turn: Draw two timelines and connect the matching time.
Explain why a shared location alone is insufficient.

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Marzano check - circle: ① ② ③

03 Look and explain

Object A reaches a crossing in 6 months. Earth reaches it in 5 months and again 15 months from now. Does crossing the same point establish an encounter?

Sentence starter: I notice ... . This supports ... because ... .

Body	Crossing time
A	6 months
Earth	3, 15 months

Marzano check - circle: ① ② ③

04 Build the solution

A toy orbit repeats every 8 years. An object was at marker P in year 0.
Name its next two returns toP. Does that tell where Earth is?

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

A student turns a ±2-day timing range into “a 50% chance of impact.” Explain why that does not follow.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

A 4 kg object is in a field of 6 N/kg.
Give its mass and weight, and distinguish the two quantities.

Word help: weight = the gravitational force on an object

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

07 Exit check

Name two pieces of information, beyond period, needed to assess a possible encounter.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Create two repeating crossing schedules that share a location but never share a time in their first 12 months.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3